DPrivBench Accepted Positive Questions

Questions, proofs, and source metadata

Run: `positive-20260729-001`
Records: 10 accepted positive questions
Sources: 5 literature PDFs
Verification: Blind independent solution followed by full proof audit

Automated workflow status: 10 ACCEPT, 0 REVISE, 0 REJECT

Generated from the accepted export and its locked provenance artifact.
A human spot check is recommended before public benchmark release.

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1 Question 1: Privacy profile bound for noisy argmax

Source: Koskela, Redberg, and Wang (2024), *Privacy Profiles for Private Selection*
Source location: Section 3 **Subject:** Private Selection
Topic: Privacy Profiles

Question

Let $X \sim X'$ denote neighboring datasets under a fixed add/remove or replace-one adjacency relation. Let $m \in \mathbb{N}$, $[m] = \{1, \dots, m\}$, and let f_i be real-valued scores satisfying $|f_i(X) - f_i(X')| \le 1$ for every $i \in [m]$ and every $X \sim X'$. Let $Z_1, \dots, Z_m$ be i.i.d. real noises with common density p, and define $M(X) = \arg\max_{i \in [m]} \{f_i(X) + Z_i\}$ using fixed data-independent tie-breaking. For distributions P, Q with densities p_P, p_Q, define the hockey-stick divergence $H_{e^t}(P\|Q) = \int [p_P(y)/p_Q(y) - e^t]_+ p_Q(y)\, dy$, where $[u]_+ = \max\{u, 0\}$. Define $\delta_2(t) = \sup_{|a-a'| \le 2} H_{e^t}(\mathrm{Law}(a + Z_1) \| \mathrm{Law}(a' + Z_1))$. Is it true that, for every $t \in \mathbb{R}$ and every $X \sim X'$, one has $H_{e^t}(M(X)\|M(X')) \le m\, \delta_2(t)$?

Verified proof

Fix $X \sim X'$ and an output index i. Conditional on $Z_i = r - 2$, if i wins on X, then i wins on X' when its noise is changed to r: indeed, $f_i(X') \ge f_i(X) - 1$ while every competing score satisfies $f_j(X') \le f_j(X) + 1$, so the extra shift by 2 preserves the winning inequality. Write $w_i(r)$ and $w_i'(r)$ for the corresponding conditional winning probabilities on X and X'. Thus the numerator probability for output i can be represented using $p(r-2)w_i(r-2)$, with $w_i(r-2) \le w_i'(r)$. For $h_t(u) = [u - e^t]_+$, convexity of its perspective and Jensen's inequality give, exactly as in the proof of Theorem 3.2, $H_{e^t}(M(X)\|M(X')) \le \sum_{i=1}^m \int h_t(p(r-2)/p(r))p(r)w_i'(r)\, dr$. Since $0 \le w_i'(r) \le 1$ and $h_t \ge 0$, this is at most $\sum_{i=1}^m \int h_t(p(r-2)/p(r))p(r)\, dr = mH_{e^t}(\mathrm{Law}(Z_1 + 2)\| \mathrm{Law}(Z_1)) \le m\delta_2(t)$. This proves the claimed bound. The source states the result as Theorem 3.2 on page 4 and gives this perspective/Jensen derivation in Appendix A.1, pages 16–17.

2 Question 2: Privacy of randomized hyperparameter tuning

Source: Koskela, Redberg, and Wang (2024), *Privacy Profiles for Private Selection*
Source location: Section 5.1 **Subject:** Private Selection
Topic: Randomized Hyperparameter Tuning

Question

Fix $\gamma \in (0, 1)$ and $\eta \in (-1, \infty)$. Let $K \in \mathbb{N}$ have the truncated negative-binomial law $D_{\eta,\gamma}$ defined, for $k \in \mathbb{N}$, by $\Pr(K = k) = \frac{(1-\gamma)^k}{\gamma^{-\eta} - 1} \prod_{i=0}^{k-1} \frac{i+\eta}{i+1}$ when $\eta \ne 0$, and by $\Pr(K = k) = \frac{(1-\gamma)^k}{k \log(1/\gamma)}$ when $\eta = 0$. A selection algorithm A draws K independently, runs a base mechanism Q on the same dataset K times, and releases the run with the best value in a fixed ordered quality-score space. Under the paper's neighboring-dataset relation, suppose Q is ε-differentially private, meaning $\Pr(Q(X) \in E) \le e^\varepsilon \Pr(Q(X') \in E)$ for all neighboring X, X' and measurable E. Is A necessarily $(\eta + 2)\varepsilon$-differentially private?

Verified proof

Let $\delta_Q(s)$ be the privacy profile of Q, so pure ε-DP gives $\delta_Q(\varepsilon) = 0$. Theorem 5.2 (page 7; proved as Theorem C.1 on pages 21–22) states that, for $m = \mathbb{E}[K]$, $H_{e^t}(A(X)\|A(X')) \le m\delta_Q(\widehat{t})$, where $\widehat{t} = t-(\eta+1)\log(e^s+(1-\gamma)\delta_Q(s)/\gamma)$ for any $s \ge 0$. Set $s = \varepsilon$ and $t = (\eta+2)\varepsilon$. Since $\delta_Q(\varepsilon) = 0$, $\widehat{t} = (\eta+2)\varepsilon - (\eta+1)\varepsilon = \varepsilon$. Hence $H_{e^{(\eta+2)\varepsilon}}(A(X)\|A(X')) \le m\delta_Q(\varepsilon) = 0$. Applying the same bound to the ordered pair X', X yields the reverse inequality, so the two output laws satisfy pure $(\eta+2)\varepsilon$-DP. This is precisely the one-step instantiation given in Corollary 5.3 and its proof on pages 7 and 22–23.

3 Question 3: Coordinate error of correlated Gaussian noise

Source: Lebeda (2025), *Better Gaussian Mechanism using Correlated Noise*
Source location: Section 3 **Subject:** Privacy Mechanisms
Topic: Correlated Gaussian Noise

Question

Let $d, n \in \mathbb{N}$, $\mu > 0$, $X = (x_1, \ldots, x_n)$ with $x_i \in [0,1]^d$, and $f(X) = \sum_{i=1}^n x_i$. Let $\mathbf{1}_d$ be the all-ones vector in $\mathbb{R}^d$. Independently sample $\eta \sim \mathcal{N}(0, (\sqrt{d}+1)/(4\mu^2))$ and $Z \sim \mathcal{N}(0, ((d+\sqrt{d})/(4\mu^2))I_d)$, where I_d is the $d \times d$ identity. If the released query vector is $\widetilde{f} = f(X)+\eta\mathbf{1}_d+Z$, is the error in every coordinate distributed as $\mathcal{N}(0, (d+2\sqrt{d}+1)/(4\mu^2))$, with standard deviation $(\sqrt{d}+1)/(2\mu)$?

Verified proof

For coordinate j, $\widetilde{f}_j - f(X)_j = \eta + Z_j$. The two summands are independent centered Gaussians, so their sum is centered Gaussian with variance equal to the sum of variances: $(\sqrt{d}+1)/(4\mu^2) + (d+\sqrt{d})/(4\mu^2) = (d+2\sqrt{d}+1)/(4\mu^2) = (\sqrt{d}+1)^2/(4\mu^2)$. Because $\mu > 0$, the standard deviation is therefore $(\sqrt{d}+1)/(2\mu)$. This is the calculation in Lemma 3.1 on page 5.

4 Question 4: GDP guarantee for the correlated mechanism

Source: Lebeda (2025), *Better Gaussian Mechanism using Correlated Noise*
Source location: Section 3 **Subject:** Privacy Mechanisms
Topic: Correlated Gaussian Noise

Question

Let $d, n \in \mathbb{N}$, $\mu > 0$, $X = (x_1, \ldots, x_n)$ with $x_i \in [0,1]^d$, $f(X) = \sum_{i=1}^n x_i$, and let $\mathbf{1}_d$ and I_d denote the all-ones vector and identity matrix. Two datasets are add/remove neighbors if one is obtained from the other by adding or removing one row. Independently sample $\eta \sim \mathcal{N}(0, (\sqrt{d}+1)/(4\mu^2))$ and $Z \sim \mathcal{N}(0, ((d+\sqrt{d})/(4\mu^2))I_d)$, and release $A(X) = (f(X) + \eta\mathbf{1}_d + Z,\ n + 2\eta)$. For distributions P, Q, let $T(P, Q)$ be their hypothesis-testing trade-off function; a mechanism is μ-GDP if $T(A(X), A(X')) \ge T(\mathcal{N}(0,1), \mathcal{N}(\mu,1))$ pointwise for

Question (continued)

every neighboring X, X'. Does A satisfy μ-GDP under add/remove adjacency?

Verified proof

Map each row x_i to $\widehat{x}_i = (2x_i - \mathbf{1}_d, d^{1/4}) \in \mathbb{R}^{d+1}$ and set $g(X) = \sum_i \widehat{x}_i$. Adding or removing one row changes g by a vector with squared norm at most $d + d^{1/2}$, so its ℓ_2 sensitivity is at most $\sqrt{d + \sqrt{d}}$. The standard Gaussian mechanism $G(X) = g(X) + W$ with $W \sim \mathcal{N}(0, ((d + \sqrt{d})/\mu^2) I_{d+1})$ is therefore μ-GDP (Lemma 2.1 combined with Lemmas 3.2–3.3). Define the bijection $h(u, v) = (2u - v\mathbf{1}_d, d^{1/4}v)$. For $A(X) = (u, v)$, its first d transformed coordinates equal $2f(X) - n\mathbf{1}_d + 2Z$, while the last equals $d^{1/4}n + 2d^{1/4}\eta$. These are the coordinates of $g(X)$ plus independent centered Gaussian noises: each has variance $(d + \sqrt{d})/\mu^2$. Hence $h(A(X))$ has exactly the law of $G(X)$. Since h is a data-independent bijection, A and G have the same trade-off functions for every neighboring pair. Thus A is μ-GDP. This is Theorem 3.1 on page 6, with the equivalence established in Lemma 3.4.

5 Question 5: Equivalence of approximate DP and trade-off DP

Source: Tran, Koskela, Räisä, and Honkela (2026), *f-Differential Privacy Filters: Validity and Approximate Solutions*
Source location: Section 2 **Subject:** Differential Privacy Foundations
Topic: f-Differential Privacy

Question

Let S, S^- be remove-neighboring datasets, and let $M(S)$ be a distribution on an output space. For distributions P, Q, define the trade-off function $T[P, Q](\alpha) = \inf_\phi\{1 - \mathbb{E}_Q[\phi] : \mathbb{E}_P[\phi] \le \alpha\}$, where the infimum ranges over measurable randomized tests ϕ taking values in $[0, 1]$. A mechanism is f-DP if both $T[M(S), M(S^-)] \ge f$ and $T[M(S^-), M(S)] \ge f$ pointwise for every remove pair. For $\varepsilon \ge 0$ and $\delta \in [0, 1]$, define $f_{\varepsilon,\delta}(\alpha) = \max\{0, 1 - \delta - e^\varepsilon\alpha, e^{-\varepsilon}(1 - \delta - \alpha)\}$. Is M (ε, δ)-DP in both directions for every remove pair if and only if it is $f_{\varepsilon,\delta}$-DP?

Verified proof

Write $P = M(S)$ and $Q = M(S^-)$. Assume first that both DP inequalities hold. For any test with $\mathbb{E}_P\phi \le \alpha$, applying $Q(A) \le e^\varepsilon P(A) + \delta$ to the randomized event selected by ϕ gives $1 - \mathbb{E}_Q\phi \ge 1 - \delta - e^\varepsilon\alpha$. Applying $P(A) \le e^\varepsilon Q(A) + \delta$ to the complementary randomized event gives $1 - \mathbb{E}_Q\phi \ge e^{-\varepsilon}(1 - \delta - \mathbb{E}_P\phi) \ge e^{-\varepsilon}(1 - \delta - \alpha)$. The type-II error is also nonnegative, so taking the infimum gives $T[P, Q] \ge f_{\varepsilon,\delta}$; swapping P, Q gives the other direction. Conversely, suppose both trade-off inequalities hold. For a measurable event E, use the deterministic test $\phi = \mathbf{1}_E$. Since this test is feasible at $\alpha = P(E)$, $1 - Q(E) \ge T[P, Q](P(E)) \ge 1 - \delta - e^\varepsilon P(E)$, hence $Q(E) \le e^\varepsilon P(E) + \delta$. Applying the same argument to $T[Q, P]$ gives $P(E) \le e^\varepsilon Q(E) + \delta$. Thus the two notions are equivalent. This is Proposition 2.6 on page 3.

6 Question 6: Tensor composition of Gaussian trade-off functions

Source: Tran, Koskela, Räisä, and Honkela (2026), *f-Differential Privacy Filters: Validity and Approximate Solutions*
Source location: Appendix A.2 **Subject:** Accounting and Composition
Topic: Gaussian Differential Privacy

Question

For distributions P, Q, let $T[P, Q]$ denote their hypothesis-testing trade-off function. For $\mu \geq 0$, define the Gaussian trade-off function $G_\mu = T[\mathcal{N}(0,1), \mathcal{N}(\mu, 1)]$. For trade-off functions represented as $f_1 = T[P_1, Q_1]$ and $f_2 = T[P_2, Q_2]$, define their tensor product by $f_1 \otimes f_2 = T[P_1 \times P_2, Q_1 \times Q_2]$. For every $\mu_1, \mu_2 \geq 0$, is $G_{\mu_1} \otimes G_{\mu_2} = G_{\sqrt{\mu_1^2+\mu_2^2}}$?

Verified proof

By the tensor definition, $G_{\mu_1} \otimes G_{\mu_2} = T[\mathcal{N}(0, I_2), \mathcal{N}((\mu_1, \mu_2), I_2)]$. Apply an orthogonal rotation sending (μ_1, μ_2) to $(r, 0)$, where $r = \sqrt{\mu_1^2 + \mu_2^2}$. This rotation is a measurable bijection and leaves the standard bivariate normal invariant, so it preserves the testing trade-off function. The rotated testing problem is $T[\mathcal{N}(0, I_2), \mathcal{N}((r, 0), I_2)]$. Its second coordinate has the same $\mathcal{N}(0, 1)$ law under both hypotheses and is independent of the first; consequently the likelihood ratio, and hence the optimal test, depends only on the first coordinate. The trade-off function therefore equals $T[\mathcal{N}(0, 1), \mathcal{N}(r, 1)] = G_r$. This is the rotation argument in Proposition A.18 on page 16.

7 Question 7: Conversion from zCDP to approximate DP

Source: Haghifam, Steinke, and Ullman (2024), *Private Geometric Median*
Source location: Section 1.4 **Subject:** Differential Privacy Foundations
Topic: Concentrated Differential Privacy

Question

Let neighboring datasets use replacement adjacency. For $\alpha > 1$, define the Renyi divergence $D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\int (dP/dQ)^\alpha\, dQ$. Suppose a randomized mechanism A is ρ-zCDP for some $\rho \geq 0$, meaning that $D_\alpha(A(X)\|A(X')) \leq \rho\alpha$ for every $\alpha > 1$ and every neighboring X, X'. For every $\delta \in (0,1)$, is A necessarily $(\rho + 2\sqrt{\rho\log(1/\delta)}, \delta)$-differentially private?

Verified proof

Fix neighboring X, X' and write $P = A(X)$, $Q = A(X')$. For any $\alpha > 1$, let $L = \log(dP/dQ)$ under P. Then $\mathbb{E}_P[e^{(\alpha-1)L}] = e^{(\alpha-1)D_\alpha(P\|Q)}$. Markov's inequality implies $P[L > s] \leq \exp((\alpha-1)(D_\alpha(P\|Q) - s))$. Taking $s = \rho\alpha + \log(1/\delta)/(\alpha-1)$ and using $D_\alpha \leq \rho\alpha$ makes this probability at most δ. For every event E, split according to $L \leq s$ and $L > s$ to obtain $P(E) \leq e^s Q(E) + \delta$. Now put $u = \alpha - 1 > 0$: $s \leq \rho + \rho u + \log(1/\delta)/u$. The last two terms are minimized at $u = \sqrt{\log(1/\delta)/\rho}$ when $\rho > 0$, yielding $s = \rho + 2\sqrt{\rho\log(1/\delta)}$; the $\rho = 0$ case follows by taking $u \to \infty$. Since the zCDP premise holds for every ordered neighboring pair, the DP inequality holds in both directions. This is Lemma 1.4 on page 5.

8 Question 8: Sensitivity of a geometric-median concentration score

Source: Haghifam, Steinke, and Ullman (2024), *Private Geometric Median*
Source location: Section 2.1 **Subject:** Private Statistics
Topic: Private Geometric Median

Question

Let $n \in \mathbb{N}$, $X = (x_1, \ldots, x_n) \in (\mathbb{R}^d)^n$, $\gamma \in [1/2, 1]$, $m = \lceil \gamma n \rceil$, and $\nu \geq 0$. Define $N_i^X(\nu) = |\{j \in [n] : \|x_j - x_i\|_2 \leq \nu\}|$ and $N_X(\nu) = \frac{1}{m}\max_{I\subseteq[n],|I|=m}\sum_{i\in I} N_i^X(\nu)$. If replacement-neighboring datasets X, X' differ in one sample, is $|N_X(\nu) - N_{X'}(\nu)| \leq 3$?

Verified proof

Relabel the differing sample as index 1. For any fixed $I \subseteq [n]$ of size m, each count $N_i(\nu)$ with $i \neq 1$ changes by at most 1, because only membership of the replaced point can change. If $1 \notin I$, the unnormalized sum changes by at most m. If $1 \in I$, the center itself changes, so $N_1(\nu)$ may change by at most n, while the remaining $m-1$ counts change by at most 1 each; the total change is at most $n+m-1$. For two indexed families $(a_I), (b_I)$, $|\max_I a_I - \max_I b_I| \leq \max_I |a_I - b_I|$. Therefore $|N_X(\nu) - N_{X'}(\nu)| \leq (n+m-1)/m \leq 1 + n/m \leq 1 + 1/\gamma \leq 3$, since $m = \lceil\gamma n\rceil \geq \gamma n$ and $\gamma \geq 1/2$. This is Lemma 2.2 on page 5 and its proof on page 24.

9 Question 9: Pure privacy of the K-norm mechanism

Source: Joseph, Kulesza, and Yu (2025), *Approximate Differential Privacy of the ℓ_2 Mechanism*
Source location: Section 2 **Subject:** Privacy Mechanisms
Topic: K-norm Mechanism

Question

Fix either add/remove or swap adjacency. Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$, let T be a statistic with sensitivity Δ, meaning $\|T(X)-T(X')\| \le \Delta$ for all neighboring X, X', and let $\sigma > 0$. Define the K-norm mechanism M to have conditional density $f_X(y) = Z^{-1}\exp(-\|y - T(X)\|/\sigma)$ on $\mathbb{R}^d$, where Z is the translation-invariant normalizing constant. Is M (Δ/σ)-differentially private?

Verified proof

For neighboring X, X' and every output y, translation invariance makes the normalizing constant identical for the two densities. Hence $f_X(y)/f_{X'}(y) = \exp((\|y - T(X')\| - \|y - T(X)\|)/\sigma)$. By the reverse triangle inequality, $\|y - T(X')\| - \|y - T(X)\| \le \|T(X) - T(X')\| \le \Delta$. Thus $f_X(y) \le e^{\Delta/\sigma} f_{X'}(y)$ pointwise. Integrating over any measurable event gives $\Pr(M(X) \in E) \le e^{\Delta/\sigma}\Pr(M(X') \in E)$; swapping the neighbors gives the reverse inequality. Therefore M is pure (Δ/σ)-DP. This is the privacy assertion of Lemma 2.4 on page 2.

10 Question 10: Mean squared error of the ℓ_2 mechanism

Source: Joseph, Kulesza, and Yu (2025), *Approximate Differential Privacy of the ℓ_2 Mechanism*
Source location: Section 4.2 **Subject:** Privacy Mechanisms
Topic: ℓ_2 Mechanism

Question

Let $d \in \mathbb{N}$ and $\sigma > 0$. The d-dimensional ℓ_2 mechanism releases $T(X) + Y$, where $Y \in \mathbb{R}^d$ has density proportional to $\exp(-\|y\|_2/\sigma)$. Is its mean squared ℓ_2 error equal to $\mathbb{E}\|Y\|_2^2 = d(d+1)\sigma^2$?

Verified proof

Use spherical coordinates and write $R = \|Y\|_2$. The angular factor cancels in normalization, so R has density $r^{d-1}e^{-r/\sigma}/(\Gamma(d)\sigma^d)$ for $r \ge 0$, i.e. a Gamma distribution with shape d and scale σ. Therefore $\mathbb{E}\|Y\|_2^2 = \mathbb{E}[R^2] = \sigma^2\Gamma(d+2)/\Gamma(d) = d(d+1)\sigma^2$. Equivalently, this is the $p = 2$ specialization of Lemma 4.1. The exact result is stated as Corollary 4.3 on page 8, and the general moment calculation is given on pages 14–15.